OxyGenius Pulse Ox Teaching Model

TA Instructions / Script:

1. **Open the box.**
   This is the OxyGenius Pulse Ox Teaching Model.
2. **Introduce the pulse oximeter.**
   "A pulse oximeter measures how much oxygen is in your blood. This is called *oxygen saturation*, or *SpO2%*. The higher the number, the more oxygen your blood is carrying, and the better."
3. **Demonstrate the real pulse oximeter.**
   - Have one student try the pulse oximeter normally
4. **Point to the finger model.**
   - The pulse oximeter finds SpO2%, or oxygen saturation, by measuring the ratio of oxygenated blood (which is carrying oxygen) to deoxygenated blood (which is not).
     i. Oxygenated blood is full of oxygen and ready to deliver it to your body. Deoxygenated blood has already dropped off its oxygen and is on its way back to the lungs.
   - Have students press buttons
     i. Red light = oxygenated blood and purple light = deoxygenated blood.
     ii. A healthy person has nearly 100% oxygen saturation, so our model shows 10 red lights (oxygenated).
     iii. With mild hypoxemia (low oxygen), you might have around 80%, so we show 8 red lights (oxygenated) and 2 purple (deoxygenated).
     iv. In a more serious case, one might have 60% oxygen saturation, so we have 6 red and 4 purple.
5. **Clarify the concept.**
   - In something like the serious case (red button), your blood isn't literally made up of 60% oxygenated and 40% deoxygenated blood in a critical condition.
     i. These lights are just a model to show the trend; less oxygen means fewer red lights, and that's how the pulse oximeter calculates your oxygen saturation, the value on the pulse oximeter screen.
     ii. Less oxygen -> less oxygenated blood and more deoxygenated blood -> lower value on the screen
6. **How does the pulse oximeter work**
   - It shines **two types of light** through your fingertip: **red light** and **infrared light**.
     i. Look at the real pulse oximeter. When it is on, you can only see the red light. Infrared is invisible to the human eye
   - Oxygenated blood absorbs more infrared light and lets red light pass through.
   - Deoxygenated blood absorbs more red light and lets infrared pass
   - By comparing how much red and infrared light comes through your finger, the pulse oximeter can estimate the ratio of oxygenated to deoxygenated blood, and from that, calculate your percent oxygen saturation, or blood oxygen level.
7. **Use the laser pointer model.**
   Explain this: To show how light behaves with different types of blood, we use a petri dish model.

It's filled with gels that represent blood — the left side is oxygenated blood (brighter red), and the right side is deoxygenated blood (darker red).

- When you shine the red laser pointer through the oxygenated side (left), more red light passes through, and you can clearly see the laser dot on the background behind it.
- Try shining the same laser on the deoxygenated side (right). Notice how less light gets through, and the laser dot on the white background is much dimmer or almost gone.
- This shows how oxygenated blood lets red light pass through easily, which is exactly what the pulse oximeter detects. That difference in light transmission is what allows it to calculate oxygen saturation.

8. **Fake Pulse Oximeter**
    - Pressing the gold circle inside the pulse oximeter activates a red light, causing the student's finger to glow. This creates a visual demonstration of how real pulse oximeters function, using red and infrared light to measure oxygen levels by detecting how much light passes through the finger. In this activity, the student's finger replaces the traditional gel laser demonstration, illustrating how varying levels of deoxygenated and oxygenated blood affect the amount of red light that is transmitted.
    - As you can see, red light easily passes through human tissue